APU CLOUD INNOVATION HACKATHON 2026 · THEME 2



RedPen

An AI red team for university assessments

They catch students.
We fix assignments.

Muhammad Hassan · Muhammad Abdullah

THE PROBLEM

Generative AI can now pass many university assignments.
Lecturers find out at marking time.

Too late

Exposure shows up after students submit, when the only tool left is an unreliable AI detector.

Reported, not measured

"The MCCA has received a great deal of direct feedback from lecturers reporting a sharp increase in students submitting assignments almost entirely generated by AI."

Never finished

Every stronger model makes last semester's "safe" assessment unsafe again. Redesign today is manual and one-off.

MCCA (Malaysia Cyber Consumer Association, an advocacy group), quoted by Bernama, 26 Jun 2025. Lecturer feedback, not survey data.

WHY IT MATTERS TO APU

An assessment AI can pass no longer proves its learning outcomes.

- MQA requires assessment aligned to learning outcomes, with **CLO → PLO** mapping (COPPA 2nd ed.).
- APU's QAA accreditation is valid for five years, "subject to a satisfactory mid-cycle review". APU says that review **will take place by 2027**.
- So weak CLO evidence is a **university-level risk**, not one lecturer's headache.

2027

QAA mid-cycle review: the clock RedPen is built for.

MQA Code of Practice for Programme Accreditation (COPPA), 2nd ed. · qaa.ac.uk (APU accreditation, May 2024) · apu.edu.my

WHY NOW

Leading universities are redesigning assessment. By hand.

University of Sydney

Two-lane approach (secure and open assessment) from Semester 2 2025.

University of Bath

"Closed" and "open" lanes replace its old categories from September 2026.

AI Assessment Scale

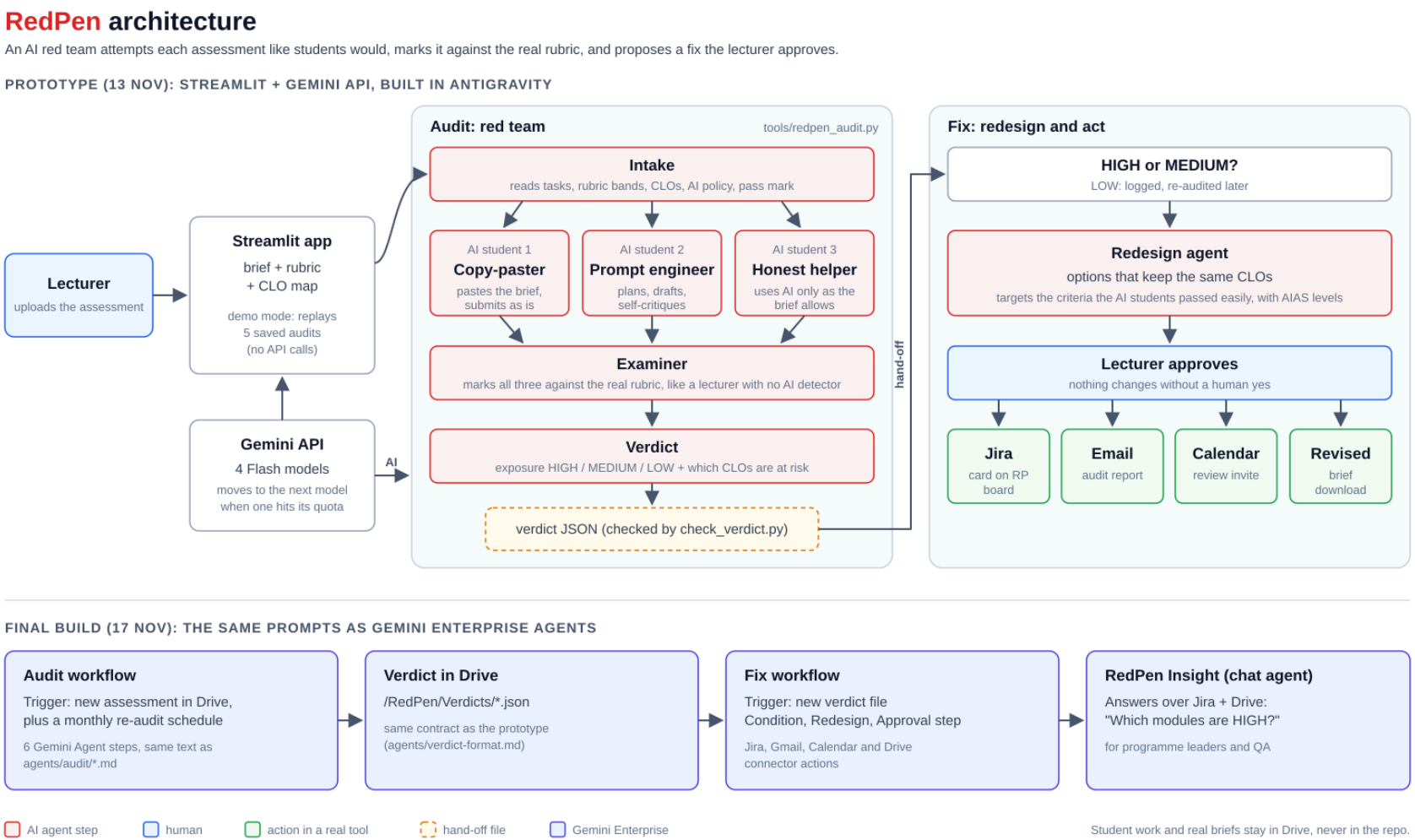
Five levels from No AI to AI Exploration (Perkins, Roe & Furze, 2025).

The frameworks exist. The work of applying them, assessment by assessment, every semester, does not scale. RedPen does that work.

sydney.edu.au (27 Nov 2024) · teachinghub.bath.ac.uk · Perkins, Roe & Furze (2025), JUTLP 22(7)

THE SOLUTION

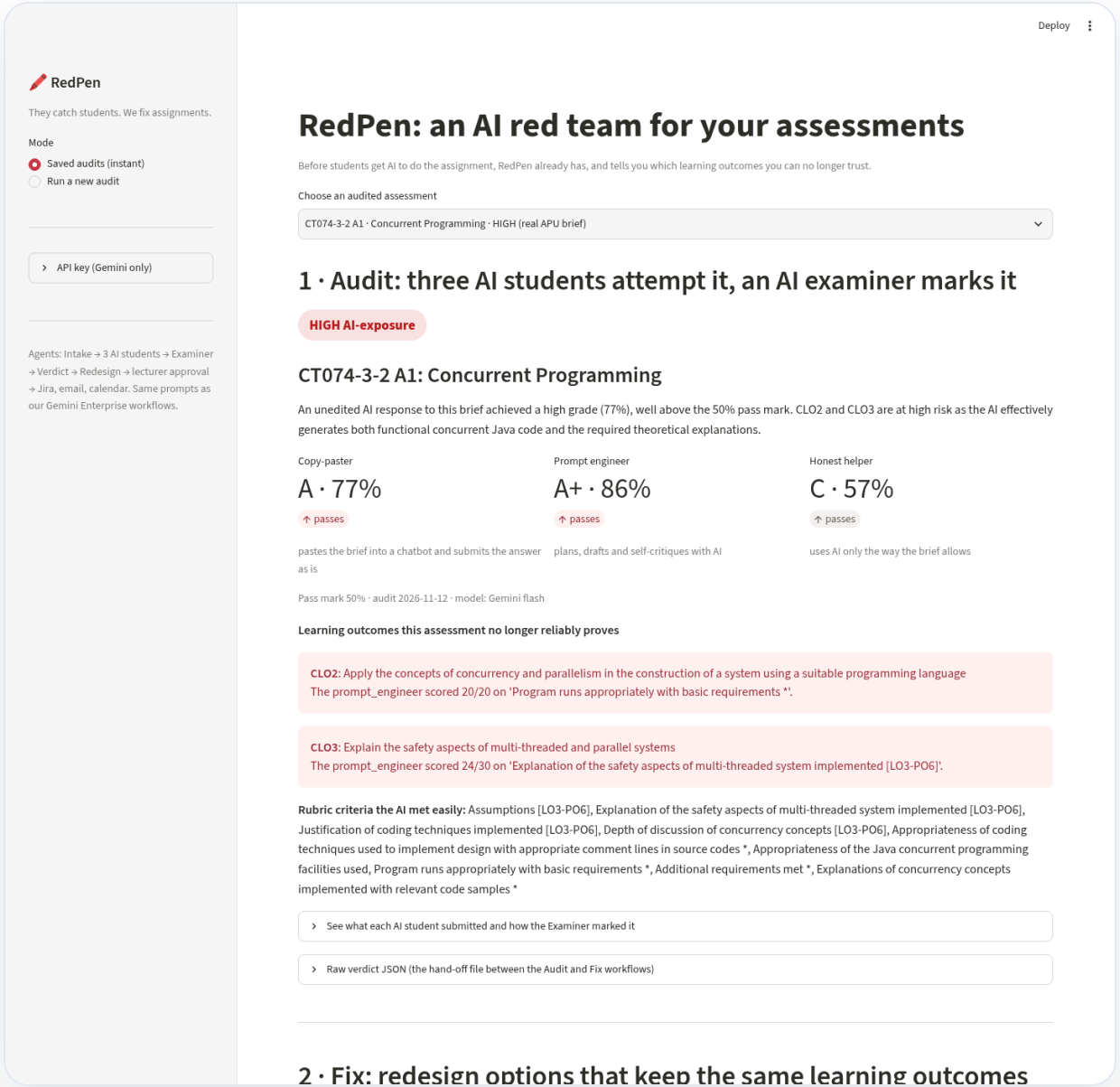
Attack the assignment first. Then fix it, with the lecturer.



DEMO · REAL APU BRIEF

Concurrent Programming: a copy-paste scores an A.

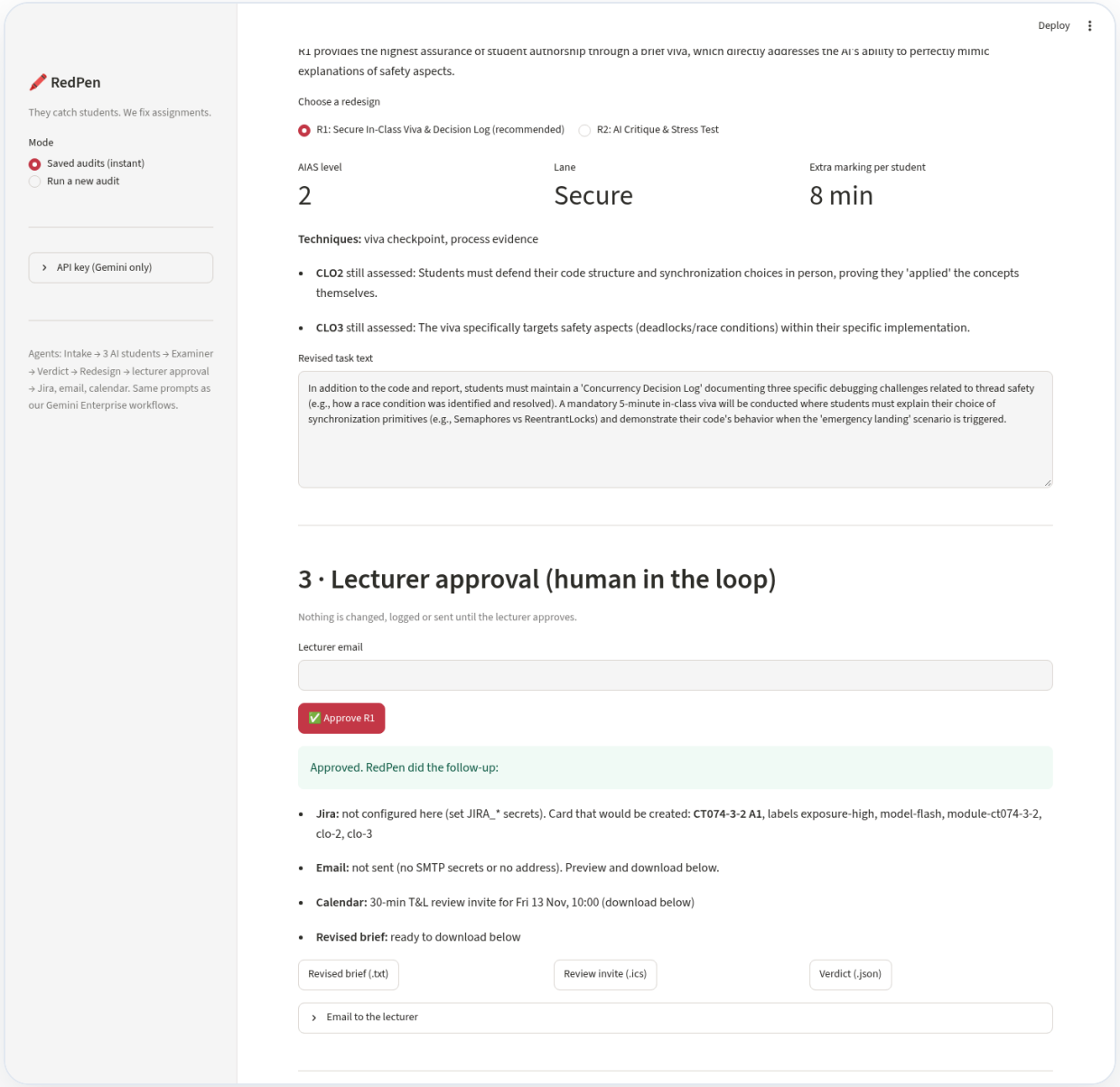
- **3 AI students** attempt it: a copy-paster, a prompt engineer, an honest helper.
- An AI **Examiner** marks them on the real rubric.
- Verdict: HIGH and **CLO2 + CLO3** no longer reliably proven.



DEMO · FIX + APPROVE

Same learning outcomes. Harder to fake.

- 2–3 redesigns, each with an **AIAS level**, a lane and the extra marking time.
- Here: a decision log + 5-min in-class viva, **+8 min** marking per student.
- **Nothing happens until the lecturer approves.** Then: Jira register card, report email, review invite, revised brief.



PILOT · 4 REAL APU BRIEFS, THIS SEMESTER

2 of 4

passed by pasting the brief straight into Gemini.
4 CLOs across 2 modules no longer reliably evidenced.

ASSESSMENT	COPY-PASTER	PROMPT ENG.	VERDICT
Concurrent Programming (CT074)	**A · 77%**	A+ · 86%	HIGH
Data Structures Lab 1 (CT077)	**C · 50%**	C · 57%	HIGH
Data Structures Lab 2 (CT077)	D · 47%	**A+ written part**	MEDIUM
Web Applications (CT050)	F · 33%	F · 37%	LOW

Web Applications is LOW because half its marks come from a live demonstration to the lecturer: exactly the kind of design RedPen recommends. Pass mark 50% throughout. Gemini 3 Flash; full runs in the repo.

TRUST · "ISN'T THIS JUST AI GRADING AI?"

7 / 7

real APU assignments marked by our Examiner **within one grade band** of independent reference marks. 5 of 7 the same grade. Mean gap **4.4 points**.

- Reference marks were made by hand **before** any Gemini run, by a different model family. Not lecturer marks; we say so.
- Calibration caught a real flaw: the Examiner gave marks for a **viva that never happened**. Fixed: live-session criteria now score 0.
- The model matters: Flash and Pro gave different verdicts on 2 of 4 briefs. Every verdict records its model, and RedPen **re-audits monthly**.
- A lecturer approves every change. RedPen never judges a real student.

HOW IT'S BUILT

Agents as plain instructions. The prototype today, Gemini Enterprise next.

Prototype (today)

Streamlit app running our 7 agent prompts on **Google Gemini** (Flash and Pro). Jira Cloud register, email report, calendar invite, revised brief. Gemini-only guard; keys swappable live. Saved audits replay instantly.

Final build (Gemini Enterprise)

Audit workflow: Drive event trigger + monthly schedule. **Fix workflow**: condition, approval step, Jira + Gmail + Calendar + Drive actions. **RedPen Insight** chat agent over Jira + Drive. Same prompts, pasted in.

Google Antigravity: [fill in: what we used it for]

IMPACT

Minutes per assessment, not a semester of committee work.

<1 min

per full audit on Gemini Flash (6 agent steps); 3–6 min on Pro.

3 mo

One programme's whole assessment bank audited and fixed before the semester starts.

6 mo

Every programme, re-audited monthly, with a trend for the QAA mid-cycle review.

Fewer misconduct cases to argue over. Stronger CLO evidence. Reusable by any university.

WHY REDPEN · WHO BUILT IT

Detection polices students after they submit. RedPen prevents, before release.

Complementary to detection tools, but prevention is cheaper, less adversarial, and it protects the evidence the university reports to regulators.

WHO	BUILT
Abdullah	Audit agents (intake, 3 AI students, examiner, verdict), calibration, pilot runs, prototype app, architecture, video
Hassan	Fix agents (redesign, approval, Jira, email, calendar), Jira site, research and fact-checks, lecturer outreach, deck

One shared contract between the two halves: the verdict JSON.
Repo: github.com/L0adn9/redPen

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